Paul Zhou

647-231-3469 | paulzhou2005@gmail.com | linkedin.com/in/chenghan-zhou | chenghan2135.github.io

EDUCATION

University of British Columbia **Sep 2023 – Present**
Bachelor of Applied Science in Mechanical Engineering, Mechatronics Option | CGPA: 83.5%

WORK EXPERIENCES

Mechanical Designer Co-op, IMW Industries Ltd., Chilliwack, British Columbia **May 2025 – Dec 2025**

- Designed and modified natural gas compressor subassemblies across 5 major customer projects using SolidWorks based on customer requests and repaired broken part and assembly files.
- Created detailed engineering drawings in SolidWorks for manufacturing and assembly with tolerances, fabrication requirements, and component information.
- Supported production staff by investigating over 30 non-conformance reports and taking necessary short-term actions and long-term design changes to address them.
- Created and maintained engineering bills of materials and product documentation using Agile PLM.

Mechanical Design Engineer, Exo Tech Society, Richmond Hill, Ontario **Jun 2024 – Aug 2024**

- Designed and manufactured a 6 DOF two-joint robotic arm with a differential wrist in SolidWorks using sheet metal and 3D printed parts.
- Performed motor sizing and torque analysis to select actuators capable of meeting payload and dynamic performance requirements.
- Evaluated mechanical concepts against weight, cost, and reliability constraints and designed using manufacturing and assembly considerations.

ENGINEERING DESIGN TEAMS

Mechanical Lead & Mechatronics Integration, UBC Thunderbots, UBC **Sep 2023 – Present**

- Led mechanical development and integrated systems for a fleet of 8 autonomous soccer-playing robots, ranking 2nd internationally in RoboCup 2024 SSL Division B.
- Supervised formal design reviews and mentored junior members in mechanical design, prototyping, design trade-offs, and mechanism reliability.
- Led the redesign of a solenoid-actuated ball-chipping mechanism, increasing chip efficiency by 58% through experimental validation, performance characterization, and iterative design.
- Collaborated with electrical and software members to resolve interdisciplinary integration challenges involving actuators, sensors, wiring, and key robot functionality including movement, dribbling, kicking, and chipping.

TECHNICAL PROJECTS

6 Degrees of Freedom Wheeled Bipedal Robot **May 2026 – Present**

- Developing a 6-actuator wheeled-legged robotic platform with independently actuated hip, knee, and wheel joints for dynamic balancing and locomotion.
- Designed quasi-direct-drive actuators using BLDC motors, magnetic absolute encoders, and custom planetary transmissions, performing motor sizing and gear-ratio analysis to meet torque and speed requirements.
- Integrated ODrive motor controllers, STM32 microcontroller, Raspberry Pi 5, and IMU to create a full-stack robotics control architecture
- Implemented field-oriented motor control with IMU feedback and inverse kinematics for joint positioning, wheel velocity.

120° Commutation Brushless DC Motor Control Simulation with MTPA and MTPV Control **May 2026 – Aug 2026**

- Developed a closed-loop PID speed control MATLAB/Simulink model of a three-phase BLDC motor using 120° electronic commutation, incorporating electrical and mechanical motor dynamics.
- Implemented PWM and dynamic switching between Maximum Torque per Ampere (MTPA) and Maximum Torque per Voltage (MTPV) control strategies to improve motor efficiency and reach target velocity under given mechanical load.
- Analyzed phase and d-q currents and voltages, electromagnetic torque, and rotor speed to characterize transient and steady-state motor behaviour and plot torque-speed characteristics.

TECHNICAL SKILLS

Robotics & Control: FOC, PID, Microcontrollers (STM32, Raspberry Pi), Kinematic modeling
Programming: Python, C, C++, MATLAB, Java, Linux, PlatformIO
Mechanical & Prototyping: SolidWorks, Autodesk Inventor, Onshape, FDM 3D printing, DFMA, FEA